Initial Investment Decision

Historical decision date and information cutoff: **31 December 2022** · USD reporting · Policy frozen before the separate evaluation period

> **Decision.** Fund both contractual payments first, then invest the residual in a constrained, long-only growth portfolio using **Utility_SingleIndex**. Do not borrow, short-sell or use derivatives.

$12,000,000	$5,235,823	$6,757,420	$6,757
INITIAL CAPITAL	LIABILITY RESERVE	GROWTH INVESTED NOTIONAL	INITIAL GROWTH TRADING COST

Full initial dollar allocation

Contractual liabilities

Purpose / instrument	Initial USD
2024 scholarship payment · Beta 2Y zero-coupon	$2,214,672
2027 learning-centre payment · Delta; coupons ring-fenced	$3,021,151
Total liability reserve	**$5,235,823**

Delta coupon reinvestment is assumed to be 0% in the base funding plan. Case bonds are treated as default-free, with executable case quotes.

Implementation and control

Growth trading cost: **10 bps one-way** on invested notional, charged upfront ($6,757). Total capital use, including that cost, is **$12,000,000** using unrounded frozen amounts.

Policy is long-only and fully invested within the growth sleeve. Maximum single common-stock weight: **15%**; maximum ETF weight: **35%**. Fractional holdings are allowed.

Growth sleeve · frozen holdings

Security	USD	Security	USD
AAPL	$206,872	MSFT	$267,237
JPM	$443,100	JNJ	$363,596
PG	$140,738	HD	$294,255
CAT	$155,145	CVX	$202,427
KO	$160,319	WMT	$236,398
SPY	$2,365,097	IWM	$1,922,235

10 U.S. common stocks across multiple sectors plus two U.S. equity ETFs. USD amounts are rounded to the nearest dollar; the frozen JSON contains full precision.

Universe: AAPL, MSFT, JPM, JNJ, PG, HD, CAT, CVX, KO, WMT, SPY and IWM.

Estimation: requested Jan 2015–Dec 2022; 95 common monthly return observations, Feb 2015–Dec 2022.

Rationale, assumptions and evaluation rule

Liabilities before growth. Match the 2024 $2.4m obligation with Beta; fund the 2027 $3.6m obligation with Delta and ring-fenced coupons. The remaining capital pursues the $8m 2030 objective in Dec-2022 purchasing power.

Growth decision. Use CAPM-style expected returns, a single-index covariance model and base risk-aversion **A=4**. Under the common comparison lens, Utility_SingleIndex has a utility score of 0.05696 versus 0.04866 for the credible rejected alternative, Eta_Historical_MaxSharpe. This is a model comparison, not a forecast or guarantee.

Material assumptions. 2.5% annual inflation is a team planning scenario, not a prediction. Compare risk aversion A=2, 4, 6 and 8. The case's “losses approaching 15%” concern has no specified metric or horizon; treat it as a diagnostic, not a hard floor.

Benchmark and frozen evaluation. SPY. Evaluate Jan 2023–Dec 2025 with target weights fixed from the 2022 decision; rebalance annually on 31 December to those weights after any scheduled liability payment. No re-estimation or re-optimization using evaluation-period data.

Open reconciliation item. At 2.5% inflation, the 2030 nominal target is $9,747,223 in the frozen policy and inflation-sensitivity output, but Section_02_Summary.csv reports $9,373,275 (the 2.0% figure). Target-dependent outputs are provisional until reconciled; this inconsistency does not alter the recorded policy allocation.

Dated Data Record

Sources and processing used to support the 31 December 2022 decision · retrieval was retrospective, after the historical information cutoff

Source inventory

Dataset / source	Retrieval and raw files	Request and settings	Use and transformation	Environment
Market prices Yahoo Finance via yfinance finance.yahoo.com	15 Oct 2026, 23:04:22 data/raw/yahoo_market_data_raw.csv Archived pickle also retained	12 tickers listed on p.1; 2015-01-01 to 2026-01-01 (end exclusive). Daily history; auto_adjust=False; actions=True; group_by=column; threads=True.	Adjusted Close; last available daily price per calendar month; simple return $P_t/P_{t-1}-1$. Monthly returns in decimal units. Common complete months only; no imputation.	yfinance 1.7.0 pandas 3.0.6 95 estimation months; 36 evaluation months
Fama–French factors Kenneth French Data Library F-F Research Data Factors	Dated_Data_Record.csv: 15 Oct 2026, 23:00:38 Raw source record: 15 Oct 2026, 21:16:53 ZIP and parsed TXT retained	Full source ZIP; monthly Mkt-RF, SMB, HML and RF rows selected; case series filtered to 2015-01 through 2025-12.	Source percentages divided by 100; YYYYMM aligned to month-end; incomplete rows excluded. Estimation through Dec-2022; Jan-2023–Dec-2025 held separate.	pandas 3.0.6 Python 3.13.15
Eligibility evidence Security eligibility as of 31 Dec 2022	data/raw/security_eligibility_2022.csv Retrieval date not documented	Historical eligibility evidence for the 12-security candidate universe.	Not a return series. The evidence file is retained and fingerprinted; the source URL and retrieval settings are not documented in the supplied record.	Retrieval package/version not documented

Integrity fingerprints · SHA-256

Raw input	SHA-256
Yahoo market CSV	`7ae950a36561fa454c8e8a1958a0ff882b244d4279eca97502c1e7c738238cf5`
French factor raw TXT	`2d0b9d3db8e03e27587fab2618648bf8ad0f8a6e8ae4826f3108832ff5ab652d`
2022 eligibility CSV	`b4d818fc54edb1bd37f4a0a5a22c3048e44ea482ba7cbad5256a3bee7ad7c157`

Timing, caveats and audit trail

Information cutoff: all estimation inputs are restricted to observations through 31 Dec 2022. The 2023–2025 data are a distinct frozen evaluation window and are not used to estimate or select the initial policy. The effective common estimation sample is Feb 2015–Dec 2022 (95 monthly observations); evaluation is Jan 2023–Dec 2025 (36 months).

Retrospective provenance: Yahoo and factor data were downloaded on 15 Oct 2026, after the historical decision date. Historical factor observations may have been revised. The candidate universe/eligibility evidence is retrospective, so this is not a point-in-time live backtest and security-selection/survivorship limitations remain.

Unresolved record gaps: download issues and their resolution are marked “NOT DOCUMENTED”; this is not evidence that no failures occurred. French factor retrieval timestamps conflict between the dated record and its raw source record (23:00:38 vs 21:16:53); verify against download logs/version history. Eligibility retrieval date and source URL are also undocumented. No missing values have been invented.

Rebuild trail: raw files are under data/raw/; processed series under data/processed/; source records under data/raw/; policy snapshot: data/frozen/initial_policy_2026-10-22.json. Retrieval scripts: src/01_download_market_data.py and src/02_download_french_factors.py. The dated record is data/Dated_Data_Record.csv. Frozen checksums are recorded in the policy snapshot.

Submission check: reconcile the factor retrieval timestamp and undocumented download/eligibility provenance before claiming a fully complete audit trail. Preserve this original dated record and frozen policy; do not replace them with later evaluation-period information.